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Subject : DBMS
Course : B. Tech CSE Core

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Pg.:

ASSIGNMENT

SECTION - A

Q1) What is database management system? Why is it required?

Ans) A database management is software that allows users to create, store, manage, retrieve and manipulate data in a structured, efficient way. DBMS is required because :

- Reduces Data Redundancy
- Provides data security
- Data integrity
- Avoids data inconsistency
- Easy data retrieval

Q2) Why is DBMS better than the traditional file system?

Ans) - Reduces data redundancy
- Backup data and recovering
- Avoids data inconsistency
- Data sharing and concurrency control
- Better data security

Q3) What is data abstraction? Explain its level?

Ans) Data abstraction in DBMS is the process of hiding complex integral details of how data is stored and maintained and showing only the necessary information to users. Level of data abstraction :-

① Physical level (Internal level):
- The lowest level of abstraction.
- Describes how data is actually stored in memory.
- Includes details like storage structures, indexing, file organisation etc.

② Logical level (conceptual level) :
- Describes what data is stored in the database.
- Defines relationships among data
- Does not show physical storage details.

③ View level (External level) :
- The high level of abstraction
- Shows only required data to specific users.
- Different users can have different views of the same database.

Q4) What is the 3 schema architecture? Why is it important?

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Ans) The three schema architecture is database design framework that separates the database system into 3 different levels to achieve data abstraction and independence. The three schema are :

(i) Eternal schema (view level)
- Describes how individual user view the data.
- Different user can have differerent views.
- Hides the rest of the database.

(ii) Conceptual level (logical level)
- Describe the overall structure of the database.
- Includes tables, attributes, relationships and constraints.
- Does not include physical storage details.

→ It is important because : • Provides data abstraction.
• Ensures data independence • Improves security.

(iii) Internal Schema (Physical level)
- Describes how data is physically stored in memory.
- Including indexing, storage structure, file organization, etc

Q5) Why is data independence? Why is it needed?

Ans) Data independence is the ability to modify the database schema at one level without affecting the schema at the next higher level.

It is needed because : • Easier database maintenance • Flexibility
• Reduced Development cost • system stability

Q6) What is an (ER) Entity-Relationship model?

Ans) The Entity Relationship model is a high level conceptual data model used to design and represent the structure of a database visually. It was introduced by Peter chen in 1976. It describes data using :
• Entities • Attributes • Relationships

It is mainly used during the database design phase.

Q7) What are Integrity constraints? Why are they important in DBMS?

Ans) Integrity constraints are rules in a DBMS that ensures the accuracy, validity and consistency of data stored in a database. They prevent invalid data from being entered into tables.

Name : PRANAV GARG
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Telco Dt.: Pg.:

Integrity constraints are important because:

- Maintain data accuracy
- Ensure consistency
- Improve flexibility
- Prevents data corruption.

SECTION-B

Q1) Draw three schema architecture and explain each level?

Ans) (i) External Schema (view level):

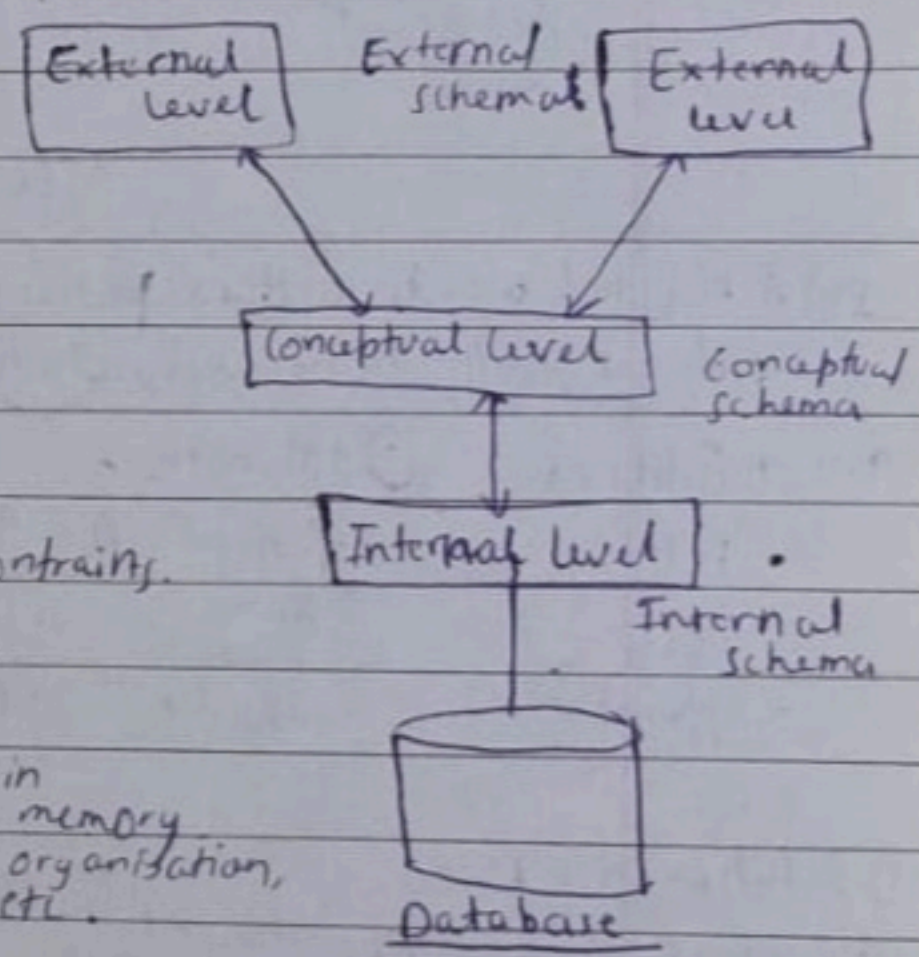
- Describes new individual users view data.
- Different users can have different views
- Hides the rest of the database.

(ii) Conceptual Schema (logical level):

- Describes the overall structure of the database.
- Includes tables, attributes, relationships, contraints.
- Does not include physical storage details.

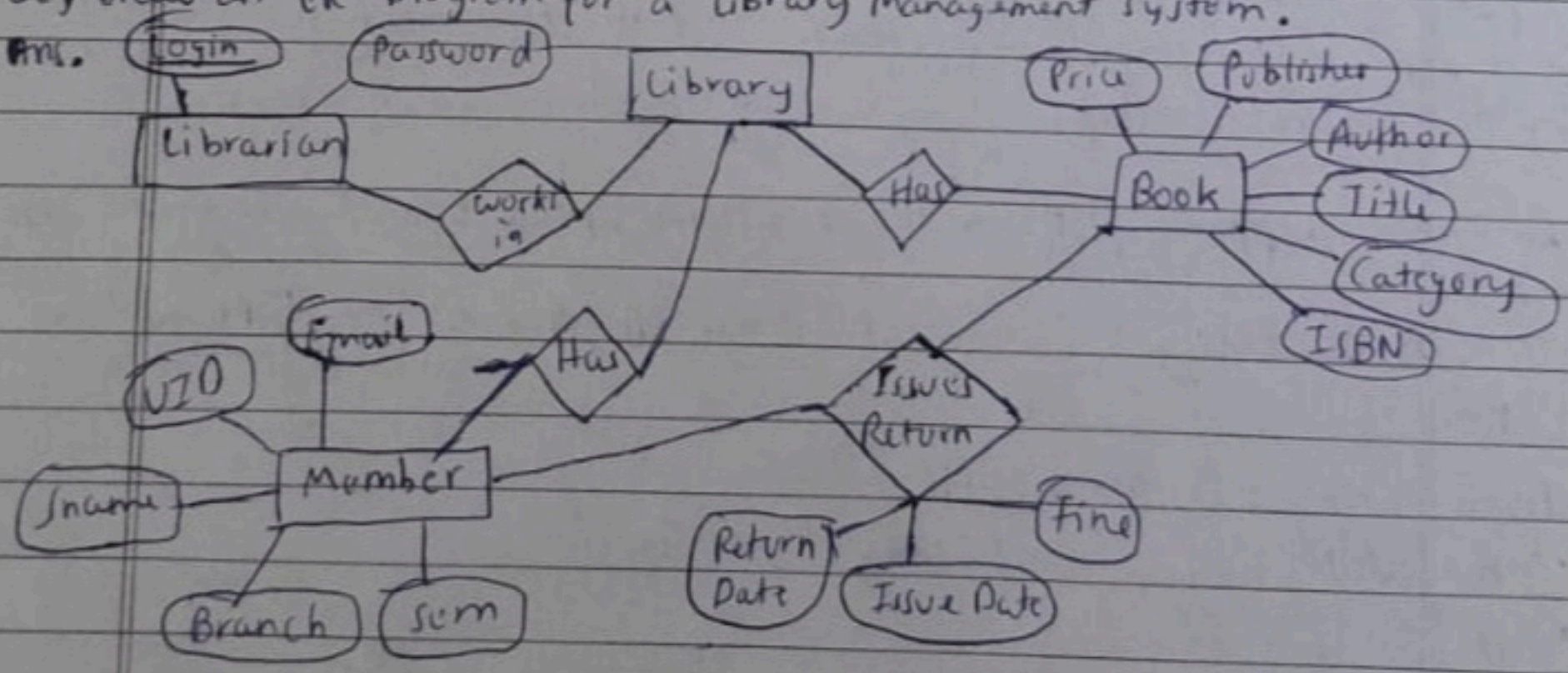
(iii) Internal Schema (Physical level):

- Describes how data is physically stored in memory
- Include indexing, storage structure, file organisation, etc.



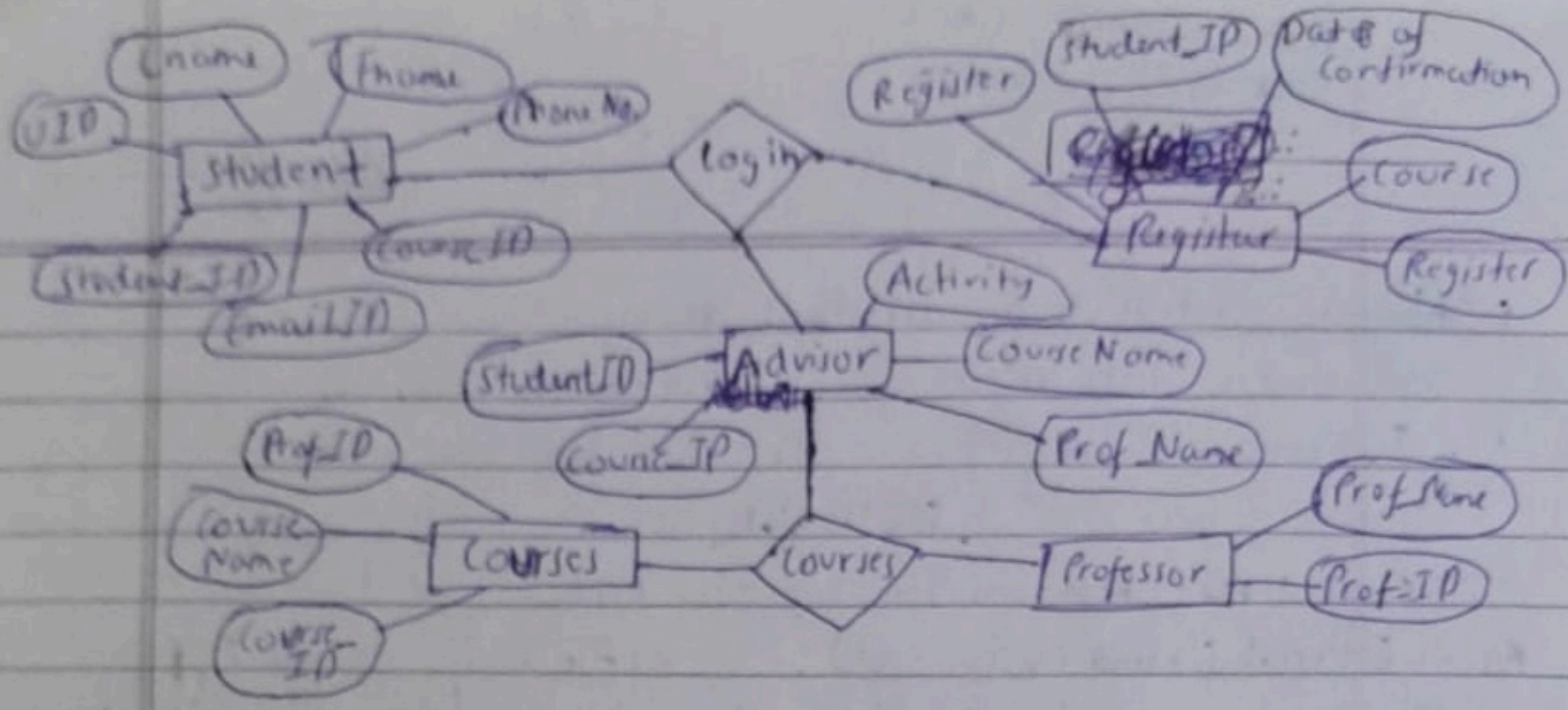
Q2) Draw an ER Diagram for a Library Management System.

Ans.



Q3) Draw an ER Diagram for a student - Course Registration System

Ans.



SECTION - C

Q1) A hospital wants to store patients, doctor, appointment data. Identify entities, attributes and relationships.

Ans ① Entities

- Patient
- Doctor
- Appointment

② Attributes :-

- Patient - ID
- Name
- Age
- Gender
- Address
- Phone-Number
- Doctor ID
- Specialisation
- Phone Number
- Experience
- Appointment - ID
- Appointment - Date
- Appointment - Time
- Status

③ Relationships:

① Patient - Books - Appointment

- A patient can book many appointments
- Each appointment belongs to one patient
- One to many relationships.

② Doctor - attends - Appointment

- A doctor can attend many appointment.
- Each appointment is handled by one doctor.
- One to many relationships.

Q2) An online shopping platform stores customer, product and order data suggest keys and constraints.

Ans) ① Customer Entity :- Customer (customer_ID, Name, Email, Phone, Address).

Keys :-

- Primary key : Customer ID
- Candidate key : Email (should be unique)
- Constraints :-
- Customer_ID → NOT NULL, UNIQUE
- ~~Customer_ID~~ Email → UNIQUE, NOT NULL
- Phone → UNIQUE (optional but recommended)
- Name → NOT NULL

NAME: PRANAV GARG
ROLL NO.: 2401010034

Telco Dt.: Pg.:

② PRODUCT ENTITY: Product (Product_ID, Product_Name, Price, Stock, Category)

Keys:
- Primary key: Product_ID

Constraints:-
- Product_ID: NOT NULL, UNIQUE
- PRICE: CHECK (Price > 0)
- Stock: CHECK (Stock >= 0)
- Product_Name: NOT NULL

③ Order Entity: Order (Order_ID, Order_Date, Customer_ID, Total_Amount, Status)

Keys:
- Primary key: Order_ID
- Foreign key: Customer_ID

Constraints:-
- Order_ID: NOT NULL, UNIQUE
- Customer_ID: FOREIGN KEY
- Total_Amount: CHECK
- Order_Date: NOT NULL
- Status: CHECK

Q3. A university stores students records. Explain how data abstraction helps different users?

Ans ① View level (External level) → For different users: At this level, each user sees only data relevant to them.

Teacher:-
- Can see: student_ID, Name, Course
- Cannot see: Fee details.

Student:
- Can see: own marks, attendance, fee status
- Cannot see: other student's records.

Accounts Department:
- Can see: Student_ID, Name, Fee Status
- Cannot see: Marks
- Benefits: Ensures security and Privacy.

② Logical level (Conceptual level):
- Defines the complete structure of the database.
- Includes tables like: Student_ID, Name, Courses, Marks, Fees.
- Managed by database design and administrators.

Benefits: Maintains overall organisation and relationships.

③ Physical level (Internal level):
- Describes how data is stored in memory.
- Includes indexing, file organisation, storage, blocks, etc.
- Handled by DBMS internally.
- Benefits: Users do not need to know how data is physically stored.

SECTION D

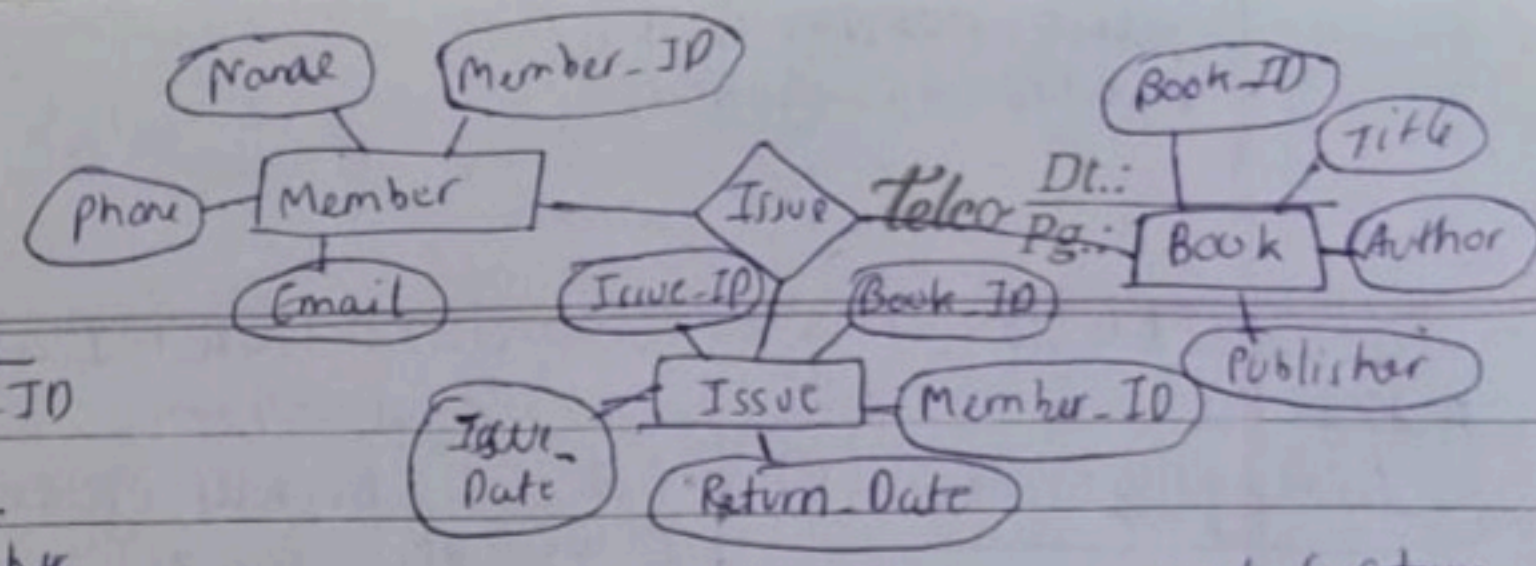
Q1) Draw a simple ER Diagram for any real life system and explain one key design decision.

Ans)

① Member
- Member_ID
- Name
- Phone
- Email

② Issue
- Issue_ID
- Issue_Date
- Return_Date
- Member_ID
- Book_ID

③ Book
- Book_ID
- Title
- Author
- Publisher



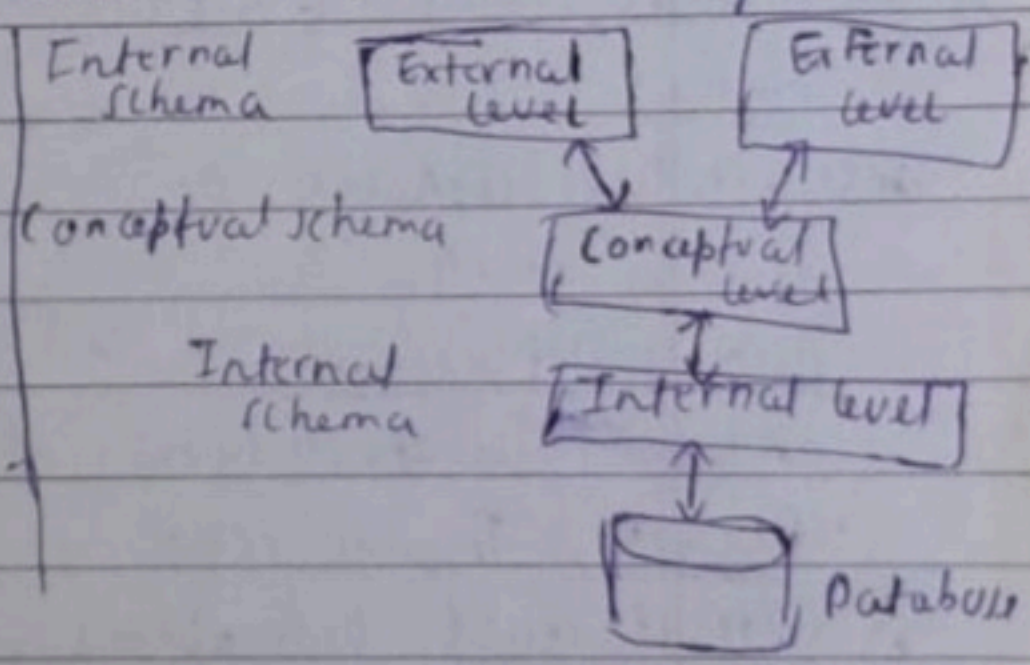
Library Management System

Relationships:

- A member can issue many books.
- A book can be issued by many members. ~~(M:N) relationships b/w member and book~~.
- This creates a many to many (M:N) relationship b/w member and book
- To resolve this, we introduce the ISSUE Entity.

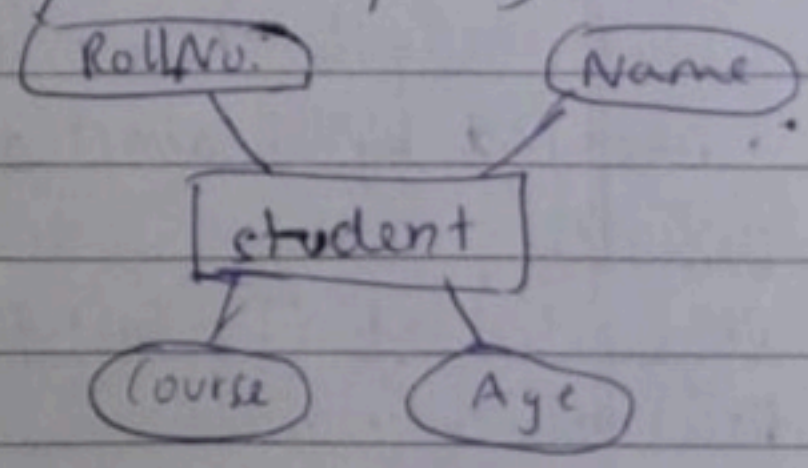
Q2) Draw the three schema Architecture and mention one benefit.

Ans) Benefit: Data Independence, changes made at the physical level (like changing storage method) do not affect the logical or external levels.



Q3) Draw an Entity with attributes and identify its Primary key:

Ans) Explanation: Entity:
Students (Represent by Rectangle)
Attributes: → Roll no.
→ Name
→ Course
→ Age



Primary key: Roll no.
- Uniquely identifies each student
- Cannot be NULL
- No two students can have the same Roll. No.